

BRYANSK, Russian Federation

WMO: 268980

Lat: 53.250N

Lon: 34.317E

Elev: 710

StdP: 14.32

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.3	-1.1	-13.1	2.8	-6.3	-6.5	4.0	0.2	21.8	30.8	19.6	27.8	4.5	20	0.536

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.2	85.2	67.6	82.1	66.0	79.1	64.3	69.5	81.0	67.8	78.3	66.2	75.9	6.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.7	97.5	73.9	64.1	92.0	72.3	62.5	86.8	71.0	34.0	81.0	32.5	78.9	31.3	75.8	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.7	16.4	14.6	-11.2	89.3	5.4	3.4	-15.1	91.8	-18.3	93.8	-21.3	95.7	-25.2	98.2		
(5)			-11.6	72.2	5.4	2.2	-15.5	73.7	-18.6	75.0	-21.6	76.2	-25.5	77.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.1	21.1	23.0	31.2	46.3	57.3	63.0	67.0	64.4	54.8	43.3	32.3	23.6		
	DBStd	18.61	10.72	10.47	8.39	8.66	8.54	6.87	5.65	5.92	7.44	7.98	9.32	10.59		
	HDD50	4063	896	755	583	172	30	2	0	0	38	236	532	819		
	HDD65	7910	1361	1175	1047	561	267	119	44	83	314	673	982	1284		
	CDD50	1893	0	0	1	61	255	392	526	448	181	28	1	0		
	CDD65	263	0	0	0	0	27	59	104	66	7	0	0	0		
	CDH74	2163	0	0	0	6	231	463	799	586	77	1	0	0		
	CDH80	500	0	0	0	0	39	93	213	151	4	0	0	0		
	WSAvg	6.5	7.0	6.9	7.7	7.2	6.6	6.5	5.4	5.6	5.7	6.4	6.7	7.0		
	PrecAvg	26.4	1.8	1.6	1.5	1.6	2.5	3.0	3.3	2.5	2.4	2.3	2.0	1.8		
(15) Precipitation	PrecMax	34.6	3.7	2.8	3.0	3.7	4.1	7.7	7.5	5.3	5.2	4.6	3.7	3.7		
	PrecMin	16.0	0.6	0.6	0.5	0.0	0.6	0.7	0.9	0.3	0.4	0.5	0.1	0.6		
	PrecStd	3.7	0.6	0.6	0.7	0.8	0.8	1.5	1.5	1.4	1.2	1.2	0.9	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.5	40.8	58.5	74.1	84.2	86.3	89.7	88.2	80.0	69.6	54.0	46.2		
		MCWB	37.6	36.8	46.9	55.5	64.2	67.8	69.1	68.3	61.8	57.8	51.0	44.8		
	2%	DB	36.8	37.7	51.6	69.0	80.3	82.6	85.6	84.1	76.4	63.2	49.2	40.1		
		MCWB	35.3	35.7	42.9	54.0	63.3	66.3	68.2	67.0	61.4	54.8	47.0	38.7		
	5%	DB	34.7	36.1	46.5	65.0	75.8	79.1	82.0	80.5	71.5	58.1	46.5	36.6		
		MCWB	33.5	34.3	39.2	51.7	61.3	64.3	67.0	65.2	59.5	52.3	44.6	35.5		
	10%	DB	33.1	34.8	42.4	60.9	71.9	76.0	78.5	76.8	66.6	54.3	43.7	34.5		
		MCWB	32.1	33.3	37.2	49.7	58.5	62.9	65.6	63.5	57.2	50.0	42.2	33.7		
	0.4%	WB	38.5	38.3	47.7	57.8	67.1	70.1	72.0	71.1	65.2	60.0	51.3	45.0		
		MCDB	39.1	39.4	57.6	69.4	79.6	82.1	84.3	83.1	76.2	66.2	52.9	45.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.3	36.4	43.4	54.8	64.5	67.9	70.1	68.8	62.4	55.8	47.8	38.6		
		MCDB	36.1	37.4	50.0	65.9	76.7	79.0	81.6	80.5	71.8	61.6	48.6	39.7		
	5%	WB	33.5	34.8	40.2	52.9	62.4	66.2	68.3	66.7	60.3	53.1	44.9	35.7		
		MCDB	34.3	35.7	44.5	63.1	73.5	76.6	78.4	77.4	68.2	57.3	46.1	36.3		
	10%	WB	32.1	33.5	37.6	50.9	60.3	64.3	66.6	64.7	58.4	50.8	42.0	33.8		
		MCDB	32.9	34.4	41.4	59.4	69.9	73.4	75.6	73.8	65.3	54.0	43.3	34.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.6	10.3	12.9	16.9	18.6	17.6	17.2	18.5	15.9	12.4	7.8	7.8		
		MCDBR	6.4	8.5	19.0	22.6	23.0	21.8	22.0	24.4	23.1	18.8	10.4	7.2		
		MCWBR	6.4	7.7	11.6	12.1	10.9	9.7	9.5	10.6	11.3	11.8	8.9	6.9		
		MCDBR	6.0	7.6	17.1	20.3	21.2	19.8	19.0	21.8	19.8	16.7	9.2	7.0		
	5% WB	MCWBR	6.0	7.3	11.2	11.4	10.8	9.6	9.3	10.5	10.7	11.5	8.7	6.9		
(40) Clear-Sky Solar Irradiance	taub		0.294	0.329	0.352	0.421	0.393	0.380	0.415	0.423	0.385	0.354	0.331	0.298		
	taud		2.135	2.064	2.122	2.137	2.313	2.370	2.291	2.252	2.327	2.376	2.257	2.159		
	Ebn at Noon		207	235	259	255	272	276	263	253	248	231	193	180		
	Edh at Noon		22	33	38	43	38	36	39	38	31	24	20	18		
(44) All-Sky Solar Radiation	RadAvg		238	542	988	1291	1744	1832	1747	1495	976	488	196	143		
	RadStd		48	76	145	111	127	158	143	158	107	87	39	37		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.17	N/A	N/A	N/A	N/A	N/A	-319	-382	N/A	N/A
(47)	Regional (7 neighbors)	+1.17	N/A	N/A	N/A	N/A	N/A	N/A	-353	+164	N/A

Nomenclature: See separate page